Requirements – 7/8 GHz Microwave Communications Equipment

This document describes the requirements for microwave radio communications equipment used for real-time seismic data telemetry. This communications equipment will transmit digitized seismic data from remote collection points to the primary and secondary data processing centers. Data must be transmitted with minimal latency for rapid data processing. The target frequencies are in the 7/8 GHz NTIA-licensed band (7.125 – 8.500 GHz).

Power

- Power supply options shall include: +24 VDC, -48 VDC, or AC (via external converter)
- Power consumption shall range from 50W to 150W per indoor unit
- The power input shall be protected from reverse polarity and overvoltage and shall support redundant power supplies

Radio Frequency

- The supported frequency bands must include: 7 GHz (7.125–7.750 GHz) and 8 GHz (7.750–8.500 GHz)
- The channel bandwidths shall be:
  - 7 GHz: 150, 175, 300 MHz
  - 8 GHz: 300, 360 MHz
- The supported modulation schemes (fixed and adapted) shall include: QPSK, 16-QAM, 32-QAM, 64-QAM, 256-QAM, 512-QAM, 1024-QAM, 2048-QAM, 4096-QAM

Data Transmission

- The radio link shall support 10–940 Mbps of throughput per link and up to 5.64 Gbps throughput per node
- The radio link shall support TDM to include DS1, DS3, and OC-3
- The supported frame size shall be up to 10,000 bytes (bidirectional)
- The radio link aggregation shall be IEEE 802.1AX LAG and proprietary L1LA

Physical/Environmental

- Operating temperature range shall be 14ºF to 131ºF
- Operating humidity range shall be 0 – 95%, non-condensing

- Connection points shall be SMA female connectors
- All connectors shall be threaded/lockable to prevent accidental disconnection

Networking/SOH

- Radios must be IP addressable and support the following protocols: IPv4 and IPv6, VLAN (IEEE 802.1Q, 802.1ad), Flow Control (IEEE 802.3x), and OAM (IEEE 802.1ag / ITU-T Y.1731)
- User interface must be web based with an HTTPS option and password protected
- Support the SNMPv3 protocol for sending SOH parameters
- Allow for remote “over the air” access to the user interface and allow users to adjust all software configurable settings
- Allow local and remote “over the air” firmware updates at no additional cost
- Allow a “back door” connection such as ssh
- Show common state-of-health (SOH), including but not limited to:

    - Signal strength (RSSI)
    - Noise level
    - VSWR (or reflected power)
    - Number of packets sent/received
    - Packet transmit/receive quality (%)
    - Input voltage
    - Board temperature
    - History plots for signal and noise levels